

The Shallows: What the Internet Is Doing to Our Brains

Nicholas Carr (2014)

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What both enthusiast and skeptic miss is what McLuhan saw: that in the long run a medium's content matters less than the medium itself in influencing how we think and act. As our window onto the world, and onto ourselves, a popular medium molds what we see and how we see it—and eventually, if we use it enough, it changes who we are, as individuals and as a society. "The effects of technology do not occur at the level of opinions or concepts," wrote McLuhan. Rather, they alter "patterns of perception steadily and without any resistance."

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The computer screen bulldozes our doubts with its bounties and conveniences. It is so much our servant that it would seem churlish to notice that it is also our master.

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what the Net seems to be doing is chipping away my capacity for concentration and contemplation. Whether I'm online or not, my mind now expects to take in information the way the Net distributes it: in a swiftly moving stream of particles.

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The paradox of neuroplasticity, observes Doidge, is that, for all the mental flexibility it grants us, it can end up locking us into "rigid behaviors."³³ The chemically triggered synapses that link our neurons program us, in effect, to want to keep exercising the circuits they've formed. Once we've wired new circuitry in our brain, Doidge writes, "we long to keep it activated."

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Bad habits can be ingrained in our neurons as easily as good ones. Pascual-Leone observes that "plastic changes may not necessarily represent a behavioral gain for a given subject." In addition to being "the mechanism for development and learning," plasticity can be "a cause of pathology."

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When it comes to the quality of our thought, our neurons and synapses are entirely indifferent. The possibility of intellectual decay is inherent in the malleability of our brains.

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Every intellectual technology, to put it another way, embodies an intellectual ethic, a set of assumptions about how the human mind works or should work. The map and the clock shared a similar ethic. Both placed a new stress on measurement and abstraction, on perceiving and defining forms and processes beyond those apparent to the senses.

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Ultimately, it's an invention's intellectual ethic that has the most profound effect on us. The intellectual ethic is the message that a medium or other tool transmits into the minds and culture of its users.

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The debate between determinists and instrumentalists is an illuminating one.

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If you look at a particular technology at a particular point in time, it certainly appears that, as the instrumentalists claim, our tools are firmly under our control.

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But if you take a broader historical or social view, the claims of the determinists gain credibility. Although individuals and communities may make very different decisions about which tools they use, that doesn't mean that as a species we've had much control over the path or pace of technological progress.

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"If the experience of modern society shows us anything," observes the political scientist Langdon Winner, "it is that technologies are not merely aids to human activity, but also powerful forces acting to reshape that activity and its meaning."

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there is one thing that determinists and instrumentalists can agree on: technological advances often mark turning points in history. New tools for hunting and farming brought changes in patterns of population growth, settlement, and labor. New modes of transport led to expansions and realignments of trade and commerce. New weaponry altered the balance of power between states.

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Neuroplasticity provides the missing link to our understanding of how informational media and other intellectual technologies have exerted their influence over the development of civilization and helped to guide, at a biological level, the history of human consciousness.

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Because language is, for human beings, the primary vessel of conscious thought, particularly higher forms of thought, the technologies that restructure language tend to exert the strongest influence over our intellectual lives. As the classical scholar Walter J. Ong put it, "Technologies are not mere exterior aids but also interior transformations of consciousness, and never more than when they affect the word."¹⁸ The history of language is also a history of the mind.

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Laws, records, transactions, decisions, traditions—everything that today would be "documented"—in oral cultures had to be, as Havelock says, "composed in formulaic verse" and distributed "by being sung or chanted aloud."³⁰ The oral world of our distant ancestors may well have had emotional and intuitive depths that we can no longer appreciate.

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The written word liberated knowledge from the bounds of individual memory and freed language from the rhythmical and formulaic structures required to support memorization and recitation.

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According to one estimate, the number of books produced in the fifty years following Gutenberg's invention equaled the number produced by European scribes during the preceding thousand years.²²

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"For the medieval type of brain," writes J. Z. Young, "making true statements depended on fitting sensory experience with the symbols of religion." The letterpress changed that. "As books became common, men could look more directly at each other's observations, with a great increase in the accuracy and content of the information conveyed."³¹ Books allowed readers to compare their thoughts and experiences not just with religious precepts, whether embedded in symbols or voiced by the clergy, but with the thoughts and experiences of others.

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In one fascinating study, conducted at Washington University's Dynamic Cognition Laboratory and published in the journal *Psychological Science* in 2009, researchers used brain scans to examine what happens inside people's heads as they read fiction. They found that "readers mentally simulate each new situation encountered in a narrative. Details about actions and sensation are captured from the text and integrated with personal knowledge from past experiences.

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One of the most important lessons we've learned from the study of neuroplasticity is that the mental capacities, the very neural circuits, we develop for one purpose can be put to other uses as well. As our ancestors imbued their minds with the discipline to follow a line of argument or narrative through a succession of printed pages, they became more contemplative, reflective, and imaginative.

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What does seem to be decreasing as Net use grows is the time we spend reading print publications—particularly newspapers and magazines, but also books.

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We don't see the forest when we search the Web. We don't even see the trees. We see twigs and leaves. As companies like Google and Microsoft perfect search engines for video and audio content, more products are undergoing the fragmentation that already characterizes written works.

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Whenever we turn on our computer, we are plunged into an "ecosystem of interruption technologies," as the blogger and science fiction writer Cory Doctorow terms it.

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There's much to be said for what economists call the "unbundling" of content. It provides people with more choices and frees them from unwanted purchases. But it also illustrates and reinforces the changing patterns of media consumption promoted by the Web. As the economist Tyler Cowen says, "When access [to information] is easy, we tend to favor the short, the sweet, and the bitty."

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Dozens of studies by psychologists, neurobiologists, educators, and Web designers point to the same conclusion: when we go online, we enter an environment that promotes cursory reading, hurried and distracted thinking, and superficial learning. It's possible to think deeply while surfing the Net, just as it's possible to think shallowly while reading a book, but that's not the type of thinking the technology encourages and rewards.

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Today's teenagers typically send or receive a message every few minutes throughout their waking hours. As the psychotherapist Michael Hausauer notes, teens and other young adults have a "terrific interest in knowing what's going on in the lives of their peers, coupled with a terrific anxiety about being out of the loop." If they stop sending messages, they risk becoming invisible.

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Searching and browsing seem to "exercise" the brain in a way similar to solving crossword puzzles, says Small. But the extensive activity in the brains of surfers also points to why deep reading and other acts of sustained concentration become so difficult online. The need to evaluate links and make related navigational choices, while also processing a multiplicity of fleeting sensory stimuli, requires constant mental coordination and decision making, distracting the brain from the work of interpreting text or other information.

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In reading online, Maryanne Wolf says, we sacrifice the facility that makes deep reading possible. We revert to being "mere decoders of information." Our ability to make the rich mental connections that form when we read deeply and without distraction remains largely disengaged.

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Even though the World Wide Web has made hypertext commonplace, indeed ubiquitous, research continues to show that people who read linear text comprehend more, remember more, and learn more than those who read text peppered with links.

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Psychological research long ago proved what most of us know from experience: frequent interruptions scatter our thoughts, weaken our memory, and make us tense and anxious. The more complex the train of thought we're involved in, the greater the impairment the distractions cause.

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Navigating the Web requires a particularly intensive form of mental multitasking. In addition to flooding our working memory with information, the juggling imposes what brain scientists call "switching costs" on our cognition. Every time we shift our attention, our brain has to reorient itself, further taxing our mental resources. As Maggie Jackson explains in *Distracted*, her book on multitasking, "the brain takes time to change goals, remember the rules needed for the new task, and block out cognitive interference from the previous, still-vivid activity."

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"Our results suggest that learning facts and concepts will be worse if you learn them while you're distracted," said the lead researcher, UCLA psychologist Russell Poldrack.

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The near-continuous stream of new information pumped out by the Web also plays to our natural tendency to “vastly overvalue what happens to us right now,” as Union College psychologist Christopher Chabris explains. We crave the new even when we know that “the new is more often trivial than essential.”

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“There is absolutely no question that modern search engines and cross-referenced websites have powerfully enabled research and communication efficiencies,” he says. “There is also absolutely no question that our brains are engaged less directly and more shallowly in the synthesis of information when we use research strategies that are all about ‘efficiency,’ ‘secondary (and out-of-context) referencing,’ and ‘once over, lightly.’”

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The ability to skim text is every bit as important as the ability to read deeply. What is different, and troubling, is that skimming is becoming our dominant mode of reading. Once a means to an end, a way to identify information for deeper study, scanning is becoming an end in itself—our preferred way of gathering and making sense of information of all sorts.

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What we’re experiencing is, in a metaphorical sense, a reversal of the early trajectory of civilization: we are evolving from being cultivators of personal knowledge to being hunters and gatherers in the electronic data forest.

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THERE ARE COMPENSATIONS. Research shows that certain cognitive skills are strengthened, sometimes substantially, by our use of computers and the Net. These tend to involve lower-level, or more primitive, mental functions such as hand-eye coordination, reflex response, and the processing of visual cues.

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“every medium develops some cognitive skills at the expense of others.”

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Google’s Silicon Valley headquarters—the Googleplex—is the Internet’s high church, and the religion practiced inside its walls is Taylorism. The company, says CEO Eric Schmidt, is “founded around the science of measurement.” It is striving to “systematize everything” it does.⁴ “We try to be very data-driven, and quantify everything,”

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What Taylor did for the work of the hand, Google is doing for the work of the mind.

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In his 1993 book *Technopoly*, Neil Postman distilled the main tenets of Taylor’s system of scientific management. Taylorism, he wrote, is founded on six assumptions: “that the primary, if not the only, goal of human labor and thought is efficiency; that technical calculation is in all respects superior to human judgment; that in fact human judgment cannot be trusted, because it is plagued by laxity, ambiguity, and unnecessary complexity; that subjectivity is an obstacle to clear thinking; that what cannot be measured either does not exist or is of no value; and that the affairs of citizens are best guided and conducted by experts.”¹¹ What’s remarkable is how well Postman’s summary encapsulates Google’s own intellectual ethic. Only one tweak is required to bring it up to date. Google doesn’t believe that the affairs of citizens are best guided by experts. It believes that those affairs are best guided by software algorithms—which is exactly what Taylor would have believed had powerful digital computers been around in his day.

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When carried into the realm of the intellect, the industrial ideal of efficiency poses, as Hawthorne understood, a potentially mortal threat to the pastoral ideal of meditative thought.

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The problem today is that we’re losing our ability to strike a balance between those two very different states of mind. Mentally, we’re in perpetual locomotion.

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With its enormous ambition, its immense bankroll, and its imperialistic designs on the world of

knowledge, Google is a natural vessel for our fears as well as our hopes. “Some say Google is God,” Sergey Brin has acknowledged. “Others say Google is Satan.”⁶¹

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Google is neither God nor Satan, and if there are shadows in the Googleplex they’re no more than the delusions of grandeur. What’s disturbing about the company’s founders is not their boyish desire to create an amazingly cool machine that will be able to outthink its creators, but the pinched conception of the human mind that gives rise to such a desire.

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Erasmus’s recommendation that every reader keep a notebook of memorable quotations was widely and enthusiastically followed. Such notebooks, which came to be called “commonplace books,” or just “commonplaces,” became fixtures of Renaissance schooling. Every student kept one.⁶ By the seventeenth century, their use had spread beyond the schoolhouse. Commonplaces were viewed as necessary tools for the cultivation of an educated mind.

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Those who celebrate the “outsourcing” of memory to the Web have been misled by a metaphor. They overlook the fundamentally organic nature of biological memory. What gives real memory its richness and its character, not to mention its mystery and fragility, is its contingency. It exists in time, changing as the body changes. Indeed, the very act of recalling a memory appears to restart the entire process of consolidation, including the generation of proteins to form new synaptic terminals.

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We don’t constrain our mental powers when we store new long-term memories. We strengthen them. With each expansion of our memory comes an enlargement of our intelligence. The Web provides a convenient and compelling supplement to personal memory, but when we start using the Web as a substitute for personal memory, bypassing the inner processes of consolidation, we risk emptying our minds of their riches.

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Learning how to think’ really means learning how to exercise some control over how and what you think,” said the novelist David Foster Wallace in a commencement address at Kenyon College in 2005. “It means being conscious and aware enough to choose what you pay attention to and to choose how you construct meaning from experience.”

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What makes us most human, Weizenbaum had come to believe, is what is least computable about us—the connections between our mind and our body, the experiences that shape our memory and our thinking, our capacity for emotion and empathy. The great danger we face as we become more intimately involved with our computers—as we come to experience more of our lives through the disembodied symbols flickering across our screens—is that we’ll begin to lose our humanness, to sacrifice the very qualities that separate us from machines. The only way to avoid that fate, Weizenbaum wrote, is to have the self-awareness and the courage to refuse to delegate to computers the most human of our mental activities and intellectual pursuits, particularly “tasks that demand wisdom.”

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Every tool imposes limitations even as it opens possibilities. The more we use it, the more we mold ourselves to its form and function.

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As the computer programmer Thomas Lord has argued, software can end up turning the most intimate and personal of human activities into mindless “rituals” whose steps are “encoded in the logic of web pages.”³³ Rather than acting according to our own knowledge and intuition, we go through the motions.

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The “frenziedness of technology,” Heidegger wrote, threatens to “entrench itself everywhere.”